

WEEK 7 – STUDENT PERFORMANCE DATA CLEANING & TRANSFORMATION

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INTRODUCTION

Raw data is rarely ready for analysis. Before any chart or dashboard can be trusted, the data needs to be loaded, checked, cleaned, transformed, and organized into a suitable structure. This report documents the work completed in Week 7 using **Microsoft Power BI Desktop**. Power Query was used for data cleaning and transformation, Model View was used for creating relationships, and DAX was used for creating calculated measures.

The main case study is the **Students Performance dataset**, which contains examination scores and background information of students. The objective is to prepare the student data for analysis by creating a Student ID, Total Marks, Average, Grade, and Result columns, unpivoting subject scores, creating relationships between tables, and developing DAX measures.

Objective

To clean, transform, and prepare the **Student Performance in Exams** dataset for analysis in Power BI by creating calculated columns, classifying student performance, reshaping subject data, preparing the data model, and creating basic DAX measures.

STEP 1 – IMPORT THE DATASET

Open **Microsoft Power BI Desktop** and import the **Students Performance** dataset.

Steps:

1. Open **Power BI Desktop**.
2. Click **Home** → **Get Data**.
3. Select **Text/CSV**.
4. Choose the **StudentsPerformance.csv** file.
5. Click **Transform Data** to open Power Query Editor.
6. Check that the student performance data is loaded correctly.

Screenshot: Students Performance dataset loaded in power bi.

The screenshot shows the Power BI Desktop interface with the 'StudentsPerformance.csv' data source loaded. The data is displayed in a table with the following columns: gender, race/ethnicity, parental level of education, lunch, test preparation course, math score, reading score, and writing score. The data is truncated to the first 200 rows. The interface includes the Power BI ribbon (File, Home, Insert, Modeling, Design, View, Optimize, Help) and the Visualizations pane on the right.

gender	race/ethnicity	parental level of education	lunch	test preparation course	math score	reading score	writing score
female	group B	bachelor's degree	standard	none	72	72	74
female	group C	some college	standard	completed	69	90	88
female	group B	master's degree	standard	none	90	95	93
male	group A	associate's degree	free/reduced	none	47	57	44
male	group C	some college	standard	none	76	78	75
female	group B	associate's degree	standard	none	71	83	78
female	group B	some college	standard	completed	88	95	92
male	group B	some college	free/reduced	none	40	43	39
male	group D	high school	free/reduced	completed	64	64	67
female	group B	high school	free/reduced	none	38	60	50
male	group C	associate's degree	standard	none	58	54	52
male	group D	associate's degree	standard	none	40	52	43
female	group B	high school	standard	none	65	81	73
male	group A	some college	standard	completed	78	72	70
female	group A	master's degree	standard	none	50	53	58
female	group C	some high school	standard	none	69	75	78
male	group C	high school	standard	none	88	89	86
female	group B	some high school	free/reduced	none	18	32	28
male	group C	master's degree	free/reduced	completed	46	42	46
female	group C	associate's degree	free/reduced	none	54	58	61

STEP 2 – CLEAN AND STANDARDIZE THE DATA

1. Select the required text columns.
2. Go to **Transform** → **Format** → **Capitalize Each Word**.
3. Apply **Trim** to remove extra spaces.
4. Apply **Clean** to remove unwanted characters.
5. Check the values for consistency.

Examples include:

- Group A, Group B, etc.
- Bachelor's Degree
- Standard
- Free/Reduced
- Completed / None

These steps ensure that similar category values are treated consistently.

Screenshot: Standardized text columns.

The screenshot shows the Power Query Editor interface. The main area displays a table with the following columns: **gender**, **race/ethnicity**, **parental level of education**, **lunch**, **test preparation course**, and **math score**. Each column header has a data type icon and a status bar showing 100% Valid, 0% Error, and 0% Empty. The table contains 26 rows of data. The right-hand pane shows the 'Query Settings' for 'StudentsPerformance', including 'Properties' and 'Applied Steps'. The 'Applied Steps' list includes: Source, Promoted Headers, Changed Type, Added Index1, Renamed Columns, Reordered Columns, and Capitalized Each Word (which is currently selected).

	gender	race/ethnicity	parental level of education	lunch	test preparation course	math score
1	Female	Group B	Bachelor's Degree	Standard	None	
2	Female	Group C	Some College	Standard	Completed	
3	Female	Group B	Master's Degree	Standard	None	
4	Male	Group A	Associate's Degree	Free/Reduced	None	
5	Male	Group C	Some College	Standard	None	
6	Female	Group B	Associate's Degree	Standard	None	
7	Female	Group B	Some College	Standard	Completed	
8	Male	Group B	Some College	Free/Reduced	None	
9	Male	Group D	High School	Free/Reduced	Completed	
10	Female	Group B	High School	Free/Reduced	None	
11	Male	Group C	Associate's Degree	Standard	None	
12	Male	Group D	Associate's Degree	Standard	None	
13	Female	Group B	High School	Standard	None	
14	Male	Group A	Some College	Standard	Completed	
15	Female	Group A	Master's Degree	Standard	None	
16	Female	Group C	Some High School	Standard	None	
17	Male	Group C	High School	Standard	None	
18	Female	Group B	Some High School	Free/Reduced	None	
19	Male	Group C	Master's Degree	Free/Reduced	Completed	
20	Female	Group C	Associate's Degree	Free/Reduced	None	
21	Male	Group D	High School	Standard	None	
22	Female	Group B	Some College	Free/Reduced	Completed	
23	Male	Group D	Some College	Standard	None	
24	Female	Group C	Some High School	Standard	None	
25	Male	Group D	Bachelor's Degree	Free/Reduced	Completed	

File Home Transform Add Column View Tools Help

Group Use First Row By as Headers Count Rows Table

Table

Queries [1]

StudentsPerformance

Table.TransformColumns(#"Capitalized Each Word",{"gender", Text.Trim, type text}, {"race/ethnicity", Text.Trim, type text}, {"parental level of education", Text.Trim, type text}, {"lunch", Text.Trim, type text}, {"test preparation course", Text.Trim, type text}, {"math score", Text.Trim, type text})

	gender	race/ethnicity	parental level of education	lunch	test preparation course	math score
1	Female	Group B	Bachelor's Degree	Standard	None	
2	Female	Group C	Some College	Standard	Completed	
3	Female	Group B	Master's Degree	Standard	None	
4	Male	Group A	Associate's Degree	Free/Reduced	None	
5	Male	Group C	Some College	Standard	None	
6	Female	Group B	Associate's Degree	Standard	None	
7	Female	Group B	Some College	Standard	Completed	
8	Male	Group B	Some College	Free/Reduced	None	
9	Male	Group D	High School	Free/Reduced	Completed	
10	Female	Group B	High School	Free/Reduced	None	
11	Male	Group C	Associate's Degree	Standard	None	
12	Male	Group D	Associate's Degree	Standard	None	
13	Female	Group B	High School	Standard	None	
14	Male	Group A	Some College	Standard	Completed	
15	Female	Group A	Master's Degree	Standard	None	
16	Female	Group C	Some High School	Standard	None	
17	Male	Group C	High School	Standard	None	
18	Female	Group B	Some High School	Free/Reduced	None	
19	Male	Group C	Master's Degree	Free/Reduced	Completed	
20	Female	Group C	Associate's Degree	Free/Reduced	None	
21	Male	Group D	High School	Standard	None	
22	Female	Group B	Some College	Free/Reduced	Completed	
23	Male	Group D	Some College	Standard	None	
24	Female	Group C	Some High School	Standard	None	
25	Male	Group D	Bachelor's Degree	Free/Reduced	Completed	

Query Settings

Properties

Name: StudentsPerformance

Applied Steps

- Source
- Promoted Headers
- Changed Type
- Added Index1
- Renamed Columns
- Reordered Columns
- Capitalized Each Word
- Trimmed Text

File Home Transform Add Column View Tools Help

Group Use First Row By as Headers Count Rows Table

Table

Queries [1]

StudentsPerformance

Table.TransformColumns(#"Trimmed Text",{"gender", Text.Clean, type text}, {"race/ethnicity", Text.Clean, type text}, {"parental level of education", Text.Clean, type text}, {"lunch", Text.Clean, type text}, {"test preparation course", Text.Clean, type text}, {"math score", Text.Clean, type text})

	gender	race/ethnicity	parental level of education	lunch	test preparation course	math score
1	Female	Group B	Bachelor's Degree	Standard	None	
2	Female	Group C	Some College	Standard	Completed	
3	Female	Group B	Master's Degree	Standard	None	
4	Male	Group A	Associate's Degree	Free/Reduced	None	
5	Male	Group C	Some College	Standard	None	
6	Female	Group B	Associate's Degree	Standard	None	
7	Female	Group B	Some College	Standard	Completed	
8	Male	Group B	Some College	Free/Reduced	None	
9	Male	Group D	High School	Free/Reduced	Completed	
10	Female	Group B	High School	Free/Reduced	None	
11	Male	Group C	Associate's Degree	Standard	None	
12	Male	Group D	Associate's Degree	Standard	None	
13	Female	Group B	High School	Standard	None	
14	Male	Group A	Some College	Standard	Completed	
15	Female	Group A	Master's Degree	Standard	None	
16	Female	Group C	Some High School	Standard	None	
17	Male	Group C	High School	Standard	None	
18	Female	Group B	Some High School	Free/Reduced	None	
19	Male	Group C	Master's Degree	Free/Reduced	Completed	
20	Female	Group C	Associate's Degree	Free/Reduced	None	
21	Male	Group D	High School	Standard	None	
22	Female	Group B	Some College	Free/Reduced	Completed	
23	Male	Group D	Some College	Standard	None	
24	Female	Group C	Some High School	Standard	None	
25	Male	Group D	Bachelor's Degree	Free/Reduced	Completed	

Query Settings

Properties

Name: StudentsPerformance

Applied Steps

- Source
- Promoted Headers
- Changed Type
- Added Index1
- Renamed Columns
- Reordered Columns
- Capitalized Each Word
- Trimmed Text
- Cleaned Text

STEP 3 – CREATE STUDENT ID

1. Go to **Add Column**.
2. Select **Index Column** → **From 0**.
3. Rename the new column as **Student id**.
4. Keep the column as **Whole Number**.

The Student ID provides a unique key for each student and is later used to connect the tables.

Screenshot: Student id column.

Power Query Editor - Query: StudentsPerformance

Table: Table.AddIndexColumn(#"changed Type", "Index.1", 1, 1, Int64.Type)

	lunch	test preparation course	math score	reading score	writing score	Index.1
1	standard	none	72	72	74	1
2	standard	completed	69	90	88	2
3	standard	none	90	95	93	3
4	free/reduced	none	47	57	44	4
5	standard	none	76	78	75	5
6	standard	none	71	83	78	6
7	standard	completed	88	95	92	7
8	free/reduced	none	40	43	39	8
9	free/reduced	completed	64	64	67	9
10	free/reduced	none	38	60	50	10
11	standard	none	58	54	52	11
12	standard	none	40	52	43	12
13	standard	none	65	81	73	13
14	standard	completed	78	72	70	14
15	standard	none	50	53	58	15
16	standard	none	69	75	78	16
17	standard	none	88	89	86	17
18	free/reduced	none	18	32	28	18
19	free/reduced	completed	46	42	46	19
20	free/reduced	none	54	58	61	20
21	standard	none	66	69	63	21
22	free/reduced	completed	65	75	70	22
23	standard	none	44	54	53	23
24	standard	none	69	73	73	24
25	free/reduced	completed	74	71	80	25

Power Query Editor - Query: StudentsPerformance

Table: Table.ReorderColumns(#"Renamed Columns",{"Student ID", "gender", "race/ethnicity", "parental level of education", "lunch", "test preparation course"})

	Student ID	gender	race/ethnicity	parental level of education	lunch	test preparation course
1	1	female	group B	bachelor's degree	standard	none
2	2	female	group C	some college	standard	completed
3	3	female	group B	master's degree	standard	none
4	4	male	group A	associate's degree	free/reduced	none
5	5	male	group C	some college	standard	none
6	6	female	group B	associate's degree	standard	none
7	7	female	group B	some college	standard	completed
8	8	male	group B	some college	free/reduced	none
9	9	male	group D	high school	free/reduced	completed
10	10	female	group B	high school	free/reduced	none
11	11	male	group C	associate's degree	standard	none
12	12	male	group D	associate's degree	standard	none
13	13	female	group B	high school	standard	none
14	14	male	group A	some college	standard	completed
15	15	female	group A	master's degree	standard	none
16	16	female	group C	some high school	standard	none
17	17	male	group C	high school	standard	none
18	18	female	group B	some high school	free/reduced	none
19	19	male	group C	master's degree	free/reduced	completed
20	20	female	group C	associate's degree	free/reduced	none
21	21	male	group D	high school	standard	none
22	22	female	group B	some college	free/reduced	completed
23	23	male	group D	some college	standard	none
24	24	female	group C	some high school	standard	none
25	25	male	group D	bachelor's degree	free/reduced	completed

STEP 4 – CREATE TOTAL MARKS

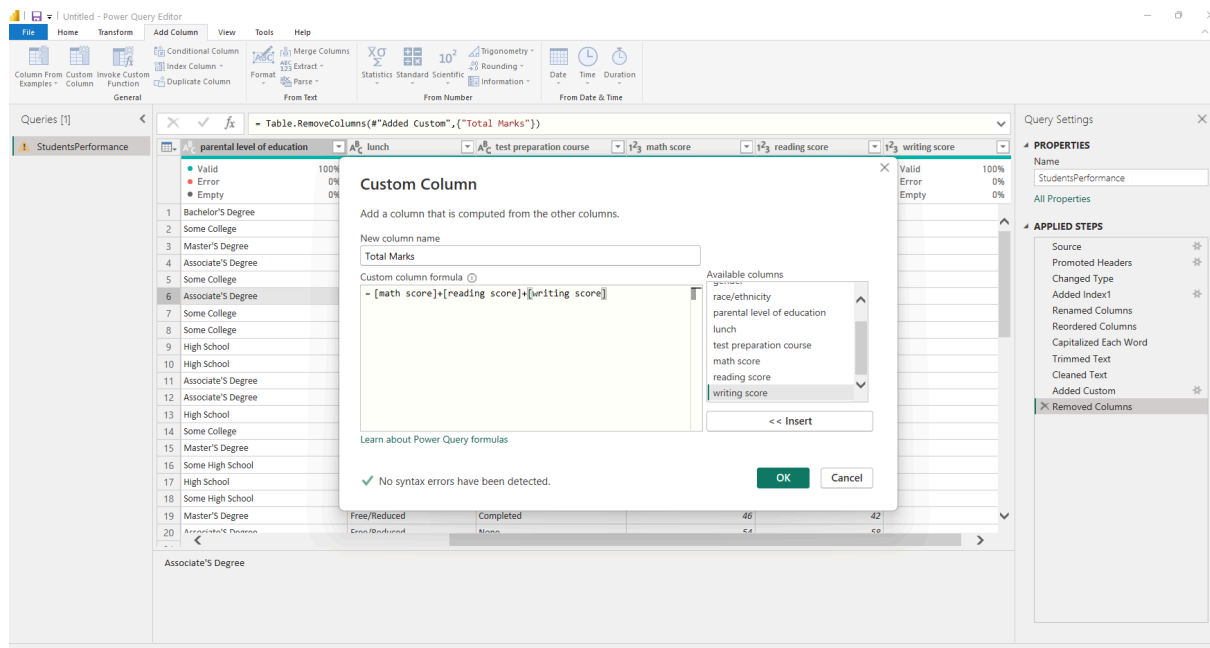
1. Go to **Add Column** → **Custom Column**.
2. Enter the column name as **Total marks**.
3. Enter the formula:

[math score] + [reading score] + [writing score]

4. Click **OK**.
5. Set **Total marks** to **Whole Number**.

The Total Marks column combines the three subject scores into one value.

Screenshot: Total marks column.



Power Query Editor - Query: StudentsPerformance

Formula Bar: `Table.AddColumn(#"Removed Columns", "Total Marks", each [math score]+[reading score]+[writing score])`

	hunch	test preparation course	math score	reading score	writing score	Total Marks
1	Standard	None	72	72	74	218
2	Standard	Completed	69	90	88	247
3	Standard	None	90	95	93	278
4	Free/Reduced	None	47	57	44	148
5	Standard	None	76	78	75	229
6	Standard	None	71	83	78	232
7	Standard	Completed	88	95	92	275
8	Free/Reduced	None	40	43	39	122
9	Free/Reduced	Completed	64	64	67	195
10	Free/Reduced	None	38	60	50	148
11	Standard	None	58	54	52	164
12	Standard	None	40	52	43	135
13	Standard	None	65	81	73	219
14	Standard	Completed	78	72	70	220
15	Standard	None	50	53	58	161
16	Standard	None	69	75	78	222
17	Standard	None	88	89	86	263
18	Free/Reduced	None	18	32	28	78
19	Free/Reduced	Completed	46	42	46	134
20	Free/Reduced	None	54	58	61	173
21	Standard	None	66	69	63	198
22	Free/Reduced	Completed	65	75	70	210
23	Standard	None	44	54	53	151
24	Standard	None	69	73	73	215
25	Free/Reduced	Completed	74	71	80	225
26						

Query Settings: StudentsPerformance

APPLIED STEPS: Source, Promoted Headers, Changed Type, Added Index1, Renamed Columns, Reordered Columns, Capitalized Each Word, Trimmed Text, Cleaned Text, Added Custom, Removed Columns, Added Custom1

STEP 5 – CREATE AVERAGE

Create the average score for each student using the three subject marks.

1. Go to **Add Column** → **Custom Column**.
2. Enter the column name as **Average**.
3. Enter the formula:

$[Total\ marks] / 3$

4. Click **OK**.
5. Check the calculated Average values.

Screenshot: Average column.

Power Query Editor interface showing the 'Custom Column' dialog box. The dialog prompts for a new column name and a custom column formula. The formula entered is $\text{Total Marks} / 3$. The background shows a table with columns: parent level of education, lunch, test preparation course, math score, reading score, and writing score. The 'Query Settings' pane on the right shows the query name 'StudentsPerformance' and the applied steps.

Query: Table.AddColumn(*Removed Columns*, "Total Marks", each [math score]+[reading score]+[writing score])

Custom Column

Add a column that is computed from the other columns.

New column name: Average

Custom column formula: $\text{Total Marks} / 3$

Available columns: parent level of education, lunch, test preparation course, math score, reading score, writing score, Total Marks

OK Cancel

Query Settings

Properties: Name: StudentsPerformance

Applied Steps: Source, Promoted Headers, Changed Type, Added Index1, Renamed Columns, Reordered Columns, Capitalized Each Word, Trimmed Text, Cleaned Text, Added Custom, Removed Columns, Added Custom1

Power Query Editor interface showing the resulting data table. The table has columns: test preparation course, math score, reading score, writing score, Total Marks, and Average. The data is sorted by the 'Average' column. The 'Query Settings' pane on the right shows the query name 'StudentsPerformance' and the applied steps.

Query: Table.AddColumn(*Added Custom1*, "Average", each [Total Marks]/3)

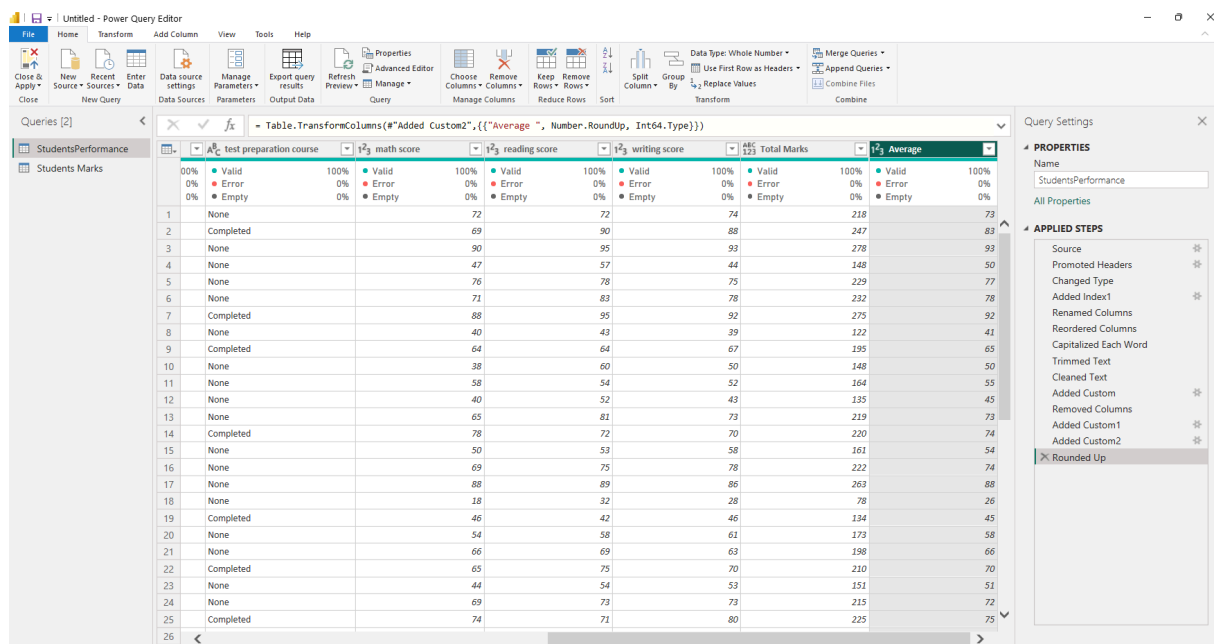
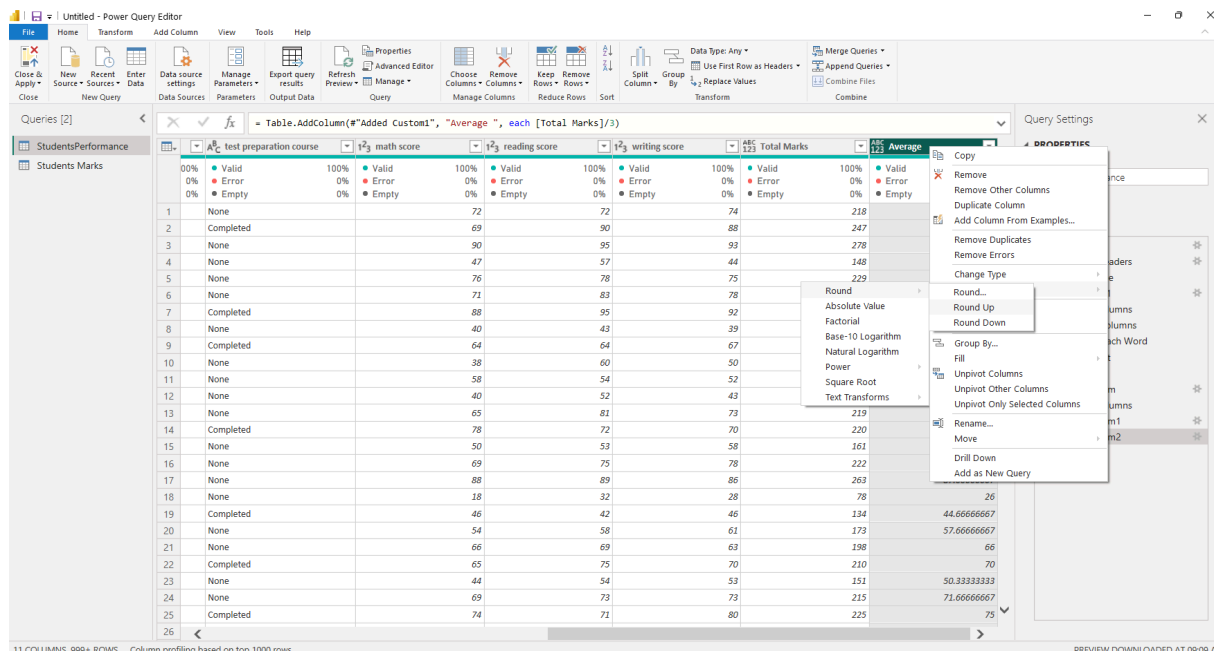
Table:

	test preparation course	math score	reading score	writing score	Total Marks	Average
1	None	72	72	74	218	72.66666667
2	Completed	69	90	88	247	82.33333333
3	None	90	95	93	278	92.66666667
4	None	47	57	44	148	49.33333333
5	None	76	78	75	229	76.33333333
6	None	71	83	78	232	77.33333333
7	Completed	88	95	92	275	91.66666667
8	None	40	43	39	122	40.66666667
9	Completed	64	64	67	195	65
10	None	38	60	50	148	49.33333333
11	None	58	54	52	164	54.66666667
12	None	40	52	43	135	45
13	None	65	81	73	219	73
14	Completed	78	72	70	220	73.33333333
15	None	50	53	58	161	53.66666667
16	None	69	75	78	222	74
17	None	88	89	86	263	87.66666667
18	None	18	32	28	78	26
19	Completed	46	42	46	134	44.66666667
20	None	54	58	61	173	57.66666667
21	None	66	69	63	198	66
22	Completed	65	75	70	210	70
23	None	44	54	53	151	50.33333333
24	None	69	73	73	215	71.66666667
25	Completed	74	71	80	225	75
26						

Query Settings

Properties: Name: StudentsPerformance

Applied Steps: Source, Promoted Headers, Changed Type, Added Index1, Renamed Columns, Reordered Columns, Capitalized Each Word, Trimmed Text, Cleaned Text, Added Custom, Removed Columns, Added Custom1, Added Custom2



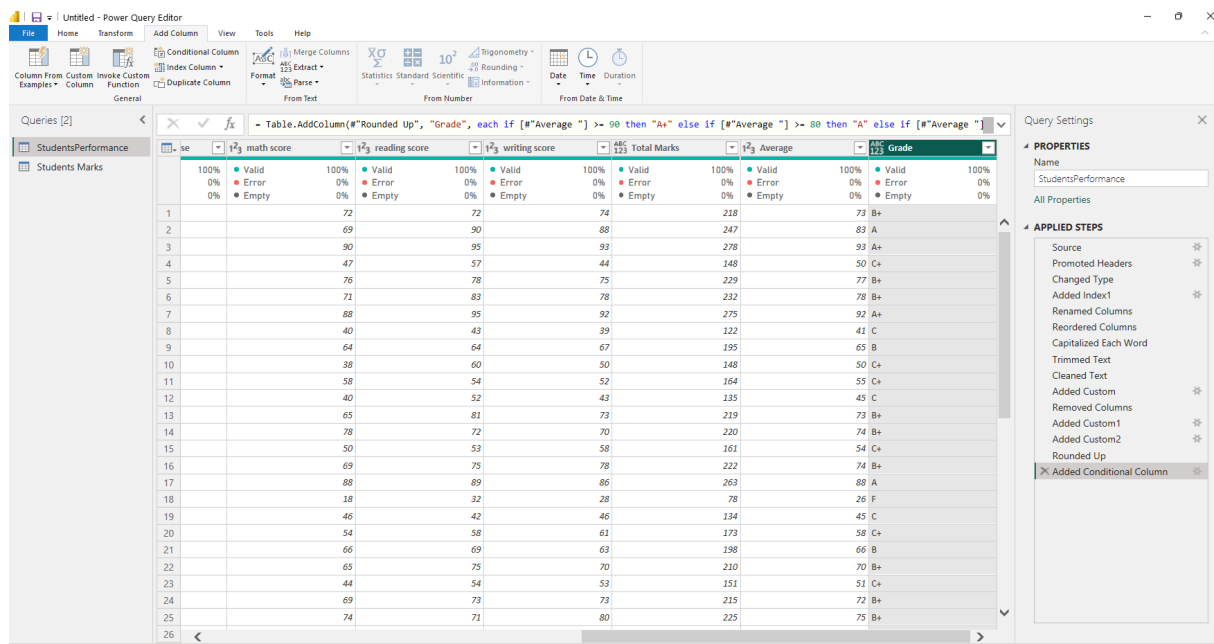
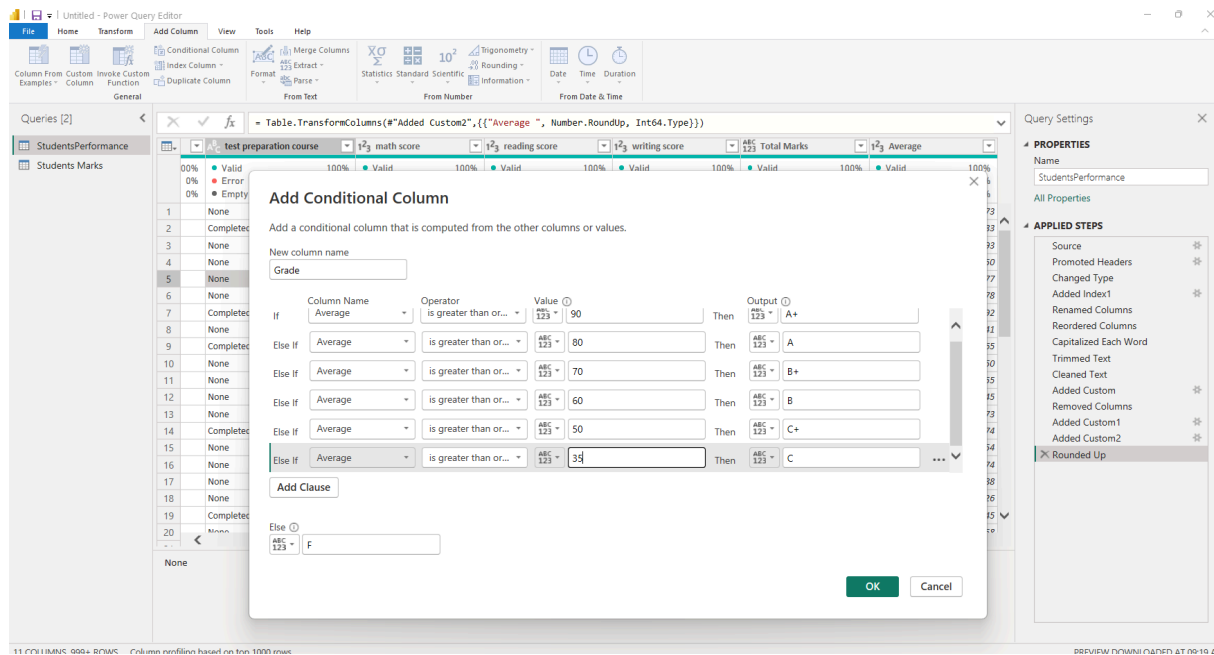
STEP 6 – CREATE GRADE

1. Go to **Add Column** → **Conditional Column**.
2. Enter the column name as **Grade**.
3. Add the grade conditions according to the grading criteria used in the dataset.

4. Click **OK**.

The Grade column assigns a grade category based on the student's performance.

Screenshot: Grade conditional column.



STEP 7 – CREATE RESULT

1. Go to **Add Column** → **Conditional Column**.
2. Enter the column name as **Result**.
3. Set the pass/fail condition.
4. In the current implementation:
 - **Grade = F** → **Fail**
 - **Otherwise** → **Pass**
5. Click **OK**.

The Result column classifies each student as **Pass** or **Fail**.

Screenshot: Result column showing Pass/Fail.

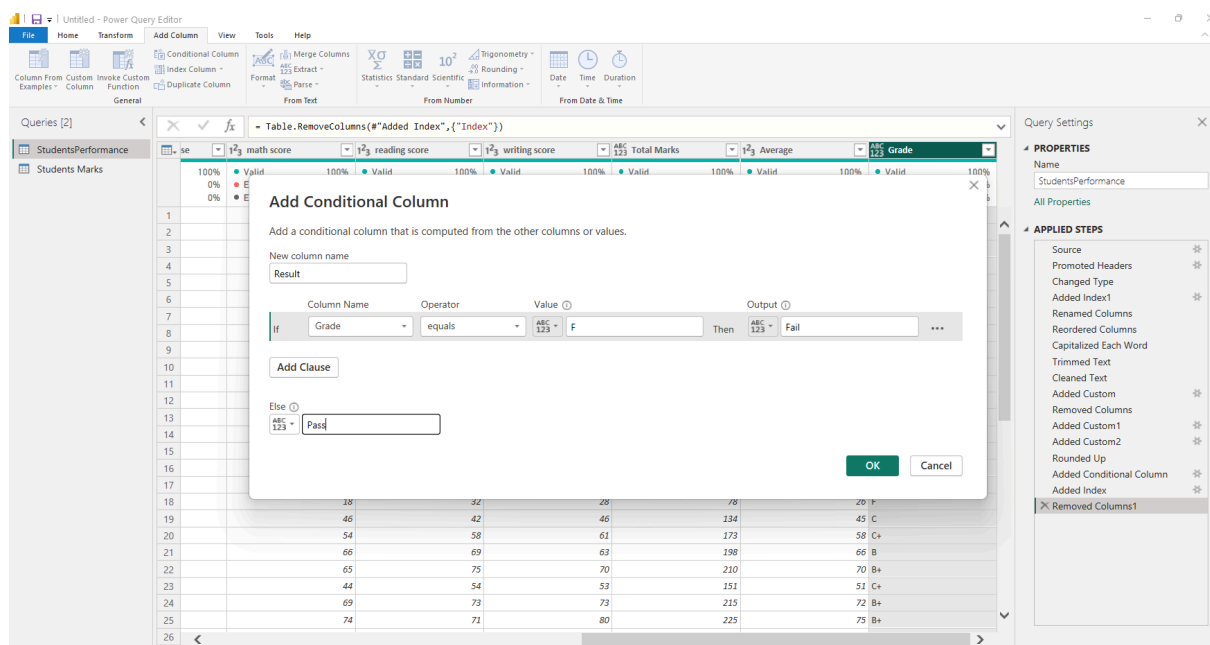


Table: AddColumn(#"Removed Columns1", "Result", each if [Grade] = "F" then "Fail" else "Pass")

	reading score	writing score	Total Marks	Average	Grade	Result
1	72	72	74	218	73 B+	Pass
2	69	90	88	247	82 A	Pass
3	90	95	93	278	93 A+	Pass
4	47	57	44	148	50 C+	Pass
5	76	78	75	229	77 B+	Pass
6	71	83	78	232	78 B+	Pass
7	88	95	92	275	92 A+	Pass
8	40	43	39	122	41 C	Pass
9	64	64	67	195	65 B	Pass
10	38	60	50	148	50 C+	Pass
11	58	54	52	164	53 C+	Pass
12	40	52	43	135	45 C	Pass
13	65	81	73	219	73 B+	Pass
14	78	72	70	220	74 B+	Pass
15	50	53	58	161	54 C+	Pass
16	69	75	78	222	74 B+	Pass
17	88	89	86	263	88 A	Pass
18	18	32	28	78	26 F	Fail
19	46	42	46	134	45 C	Pass
20	54	58	61	173	58 C+	Pass
21	66	69	63	198	66 B	Pass
22	65	75	70	210	70 B+	Pass
23	44	54	53	151	51 C+	Pass
24	69	73	73	215	72 B+	Pass
25	74	71	80	225	75 B+	Pass
26						

STEP 8 – CREATE STUDENTS MARK TABLE

The subject scores need to be converted from columns into rows using **Unpivot**.

1. Right-click the main query.
2. Select **Duplicate**.
3. Keep only:
 - Student id
 - math score
 - reading score
 - writing score
4. Select **Student id**.
5. Go to **Transform** → **Unpivot Columns** → **Unpivot Other Columns**.
6. Rename **Attribute** as **Subject**.
7. Rename **Value** as **Score**.
8. Rename the query as **Students Mark**.

This creates a long-format table where each student has one row for each subject.

Screenshot: Students Mark table showing Student id, Subject and Score.

Power Query Editor interface showing a table named 'Students Marks' with columns: Student ID, math score, reading score, and writing score. The table contains 26 rows of data. The right sidebar shows the 'Query Settings' pane with 'Properties' and 'Applied Steps'.

Student ID	math score	reading score	writing score
1	1	72	72
2	2	69	90
3	3	90	95
4	4	47	57
5	5	76	78
6	6	71	83
7	7	88	95
8	8	40	43
9	9	64	64
10	10	38	60
11	11	58	54
12	12	40	52
13	13	65	81
14	14	78	72
15	15	50	53
16	16	69	75
17	17	88	89
18	18	18	32
19	19	46	42
20	20	54	58
21	21	66	69
22	22	65	75
23	23	44	54
24	24	69	73
25	25	74	71
26	26	73	74

Power Query Editor interface showing the same table as above, but with a context menu open over the 'writing score' column header. The menu options include: Copy, Remove Columns, Remove Other Columns, Add Column From Examples..., Remove Duplicates, Remove Errors, Replace Values..., Fill, Change Type, Transform, Merge Columns, Sum, Product, Group By..., Unpivot Columns, Unpivot Only Columns, Unpivot Only Selected Columns, and Move.

Student ID	math score	reading score	writing score
1	1	72	72
2	2	69	90
3	3	90	95
4	4	47	57
5	5	76	78
6	6	71	83
7	7	88	95
8	8	40	43
9	9	64	64
10	10	38	60
11	11	58	54
12	12	40	52
13	13	65	81
14	14	78	72
15	15	50	53
16	16	69	75
17	17	88	89
18	18	18	32
19	19	46	42
20	20	54	58
21	21	66	69
22	22	65	75
23	23	44	54
24	24	69	73
25	25	74	71
26	26	73	74

Power Query Editor interface showing the transformation of a table. The main query is named "Table.UnpivotOtherColumns". The data is displayed in a table with columns: Student ID, Attribute, and Value. The table contains 26 rows of data, including scores for various subjects (math, reading, writing) for different students.

Student ID	Attribute	Value
1	math score	72
2	reading score	72
3	writing score	74
4	math score	69
5	reading score	90
6	writing score	88
7	math score	90
8	reading score	95
9	writing score	93
10	math score	47
11	reading score	57
12	writing score	44
13	math score	76
14	reading score	78
15	writing score	75
16	math score	71
17	reading score	83
18	writing score	78
19	math score	88
20	reading score	95
21	writing score	92
22	math score	40
23	reading score	43
24	writing score	39
25	math score	64
26	reading score	64

The right-hand pane shows the "Query Settings" for "Students Marks". The "APPLIED STEPS" list includes: Source, Promoted Headers, Changed Type, Added Index1, Renamed Columns, Reordered Columns, Capitalized Each Word, Trimmed Text, Cleaned Text, Added Custom, Removed Columns, Added Custom1, Added Custom2, and Removed Other Columns. The "Unpivoted Columns" step is currently selected.

Power Query Editor interface showing the transformation of a table. The main query is named "Table.RenameColumns". The data is displayed in a table with columns: Student ID, Subject, and Score. The table contains 26 rows of data, including scores for various subjects (math, reading, writing) for different students.

Student ID	Subject	Score
1	math score	72
2	reading score	72
3	writing score	74
4	math score	69
5	reading score	90
6	writing score	88
7	math score	90
8	reading score	95
9	writing score	93
10	math score	47
11	reading score	57
12	writing score	44
13	math score	76
14	reading score	78
15	writing score	75
16	math score	71
17	reading score	83
18	writing score	78
19	math score	88
20	reading score	95
21	writing score	92
22	math score	40
23	reading score	43
24	writing score	39
25	math score	64
26	reading score	64

The right-hand pane shows the "Query Settings" for "Students Marks". The "APPLIED STEPS" list includes: Source, Promoted Headers, Changed Type, Added Index1, Renamed Columns, Reordered Columns, Capitalized Each Word, Trimmed Text, Cleaned Text, Added Custom, Removed Columns, Added Custom1, Added Custom2, Removed Other Columns, Unpivoted Columns, and Renamed Columns1. The "Renamed Columns1" step is currently selected.

STEP 9 – CREATE STUDENTS DETAILS TABLE

The student-related information needs to be separated into a **Students Details** table.

1. Right-click the main query.

2. Select **Duplicate**.
3. Keep only the required student details:
 - Student id
 - Gender
 - Race/Ethnicity
 - Parental Level of Education
 - Lunch
 - Test Preparation Course
 - Average
 - Grade
 - Result
4. Remove the unnecessary columns.
5. Check that all required student details are available.
6. Rename the query as **Students Details**.

This creates a separate table containing the student's background information along with their **Average, Grade, and Result** for analysis.

Screenshot: Students Details table showing Student id, student information, Average, Grade and Result.

POWER BI WEEK 7

File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Sources Enter Data Data source settings Manage Parameters Export query results Refresh Preview Advanced Editor Choose Columns Remove Columns Keep Rows Remove Rows Sort Split Column Group By Data Type: Whole Number Use First Row as Headers Replace Values Merge Queries Append Queries Combine Files Combine

Queries [3]

StudentsPerformance

Students Marks

Students Details

Table.SelectColumns(#'Added Conditional Column1',{'Student ID', 'gender', 'race/ethnicity', 'parental level of education', 'lunch', 'test preparation course'})

	Student ID	gender	race/ethnicity	parental level of education	lunch	test preparation course	
1	1	Female	Group B	Bachelor's Degree	Standard	None	Valid 100%
2	2	Female	Group C	Some College	Standard	Completed	Valid 100%
3	3	Female	Group B	Master's Degree	Standard	None	Valid 100%
4	4	Male	Group A	Associate's Degree	Free/Reduced	None	Valid 100%
5	5	Male	Group C	Some College	Standard	None	Valid 100%
6	6	Female	Group B	Associate's Degree	Standard	None	Valid 100%
7	7	Female	Group B	Some College	Standard	Completed	Valid 100%
8	8	Male	Group B	Some College	Free/Reduced	None	Valid 100%
9	9	Male	Group D	High School	Free/Reduced	Completed	Valid 100%
10	10	Female	Group B	High School	Free/Reduced	None	Valid 100%
11	11	Male	Group C	Associate's Degree	Standard	None	Valid 100%
12	12	Male	Group D	Associate's Degree	Standard	None	Valid 100%
13	13	Female	Group B	High School	Standard	None	Valid 100%
14	14	Male	Group A	Some College	Standard	Completed	Valid 100%
15	15	Female	Group A	Master's Degree	Standard	None	Valid 100%
16	16	Female	Group C	Some High School	Standard	None	Valid 100%
17	17	Male	Group C	High School	Standard	None	Valid 100%
18	18	Female	Group B	Some High School	Free/Reduced	None	Valid 100%
19	19	Male	Group C	Master's Degree	Free/Reduced	Completed	Valid 100%
20	20	Female	Group C	Associate's Degree	Free/Reduced	None	Valid 100%
21	21	Male	Group D	High School	Standard	None	Valid 100%
22	22	Female	Group B	Some College	Free/Reduced	Completed	Valid 100%
23	23	Male	Group D	Some College	Standard	None	Valid 100%
24	24	Female	Group C	Some High School	Standard	None	Valid 100%
25	25	Male	Group D	Bachelor's Degree	Free/Reduced	Completed	Valid 100%

Query Settings

PROPERTIES

Name

Students Details

APPLIED STEPS

Source

Promoted Headers

Changed Type

Added Index1

Renamed Columns

Reordered Columns

Capitalized Each Word

Trimmed Text

Cleaned Text

Added Custom

Removed Columns

Added Custom1

Added Custom2

Rounded Up

Added Conditional Column

Added Index

Removed Columns1

Added Conditional Column1

Removed Other Columns

STEP 10 – PREPARE THE DATA MODEL

After completing the transformations:

1. Click **Home** → **Close & Apply**.
2. Go to **Model View**.
3. Check the tables:
 - StudentsPerformance
 - Students Details
 - Students Mark
 - Measure Table
4. Use **Student id** to create the required relationships.

The Student ID is used as the key to connect the student-related tables.

Screenshot: Data Model showing tables and relationships.

The screenshot displays the Power BI Desktop interface with the 'StudentsPerformance' table loaded. The table has the following columns: Student ID, gender, race/ethnicity, parental level of education, lunch, and test preparation course. The 'Query Settings' pane on the right shows the 'APPLIED STEPS' for the query, including 'Added Conditional Column1'.

Student ID	gender	race/ethnicity	parental level of education	lunch	test preparation course
1	Female	Group B	Bachelor's Degree	Standard	None
2	Female	Group C	Some College	Standard	Completed
3	Female	Group B	Master's Degree	Standard	None
4	Male	Group A	Associate's Degree	Free/Reduced	None
5	Male	Group C	Some College	Standard	None
6	Female	Group B	Associate's Degree	Standard	None
7	Female	Group B	Some College	Standard	Completed
8	Male	Group B	Some College	Free/Reduced	None
9	Male	Group D	High School	Free/Reduced	Completed
10	Female	Group B	High School	Free/Reduced	None
11	Male	Group C	Associate's Degree	Standard	None
12	Male	Group D	Associate's Degree	Standard	None
13	Female	Group B	High School	Standard	None
14	Male	Group A	Some College	Standard	Completed
15	Female	Group A	Master's Degree	Standard	None
16	Female	Group C	Some High School	Standard	None
17	Male	Group C	High School	Standard	None
18	Female	Group B	Some High School	Free/Reduced	None
19	Male	Group C	Master's Degree	Free/Reduced	Completed
20	Female	Group C	Associate's Degree	Free/Reduced	None
21	Male	Group D	High School	Standard	None
22	Female	Group B	Some College	Free/Reduced	Completed
23	Female	Group D	Some College	Standard	None
24	Female	Group C	Some High School	Standard	None
25	Male	Group D	Bachelor's Degree	Free/Reduced	Completed

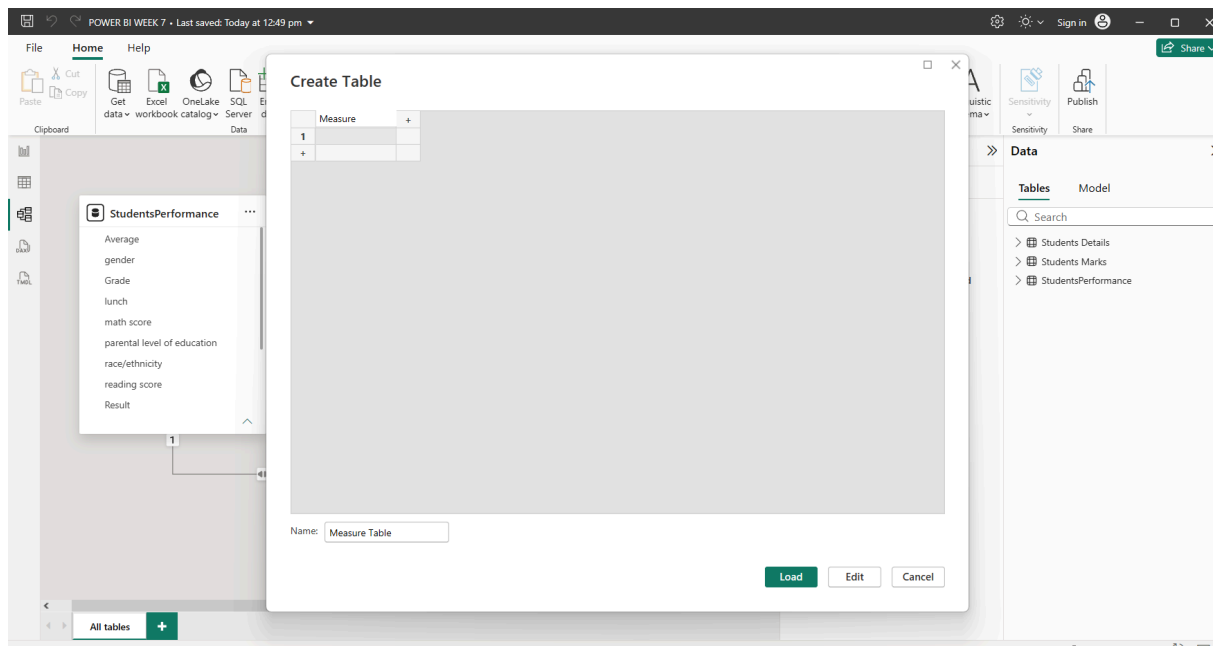
STEP 11 – CREATE MEASURE TABLE

1. Go to **Home** → **Enter Data**.
2. Create a column named **Measure**.
3. Enter a value such as **1**.

4. Name the table **Measure Table**.
5. Click **Load**.
6. Use this table to store the DAX measures.

A dedicated measure table keeps the calculations organized separately from the data columns.

Screenshot: Measure Table.



STEP 12 – CREATE DAX MEASURES

Create the following **five DAX measures**.

1. Average Math Score

Average Math Score =

AVERAGE('StudentsPerformance'[math score])

2. Average Reading Score

Average Reading Score =

AVERAGE('StudentsPerformance'[reading score])

3. Average Writing Score

Average Writing Score =

AVERAGE('StudentsPerformance'[writing score])

4. Overall Average Score

Overall Average Score =

AVERAGE('Students Details'[Average])

5. Overall Pass %

Overall Pass % =

DIVIDE(

CALCULATE(

COUNT('Students Details'[Result]),

'Students Details'[Result] = "Pass"

),

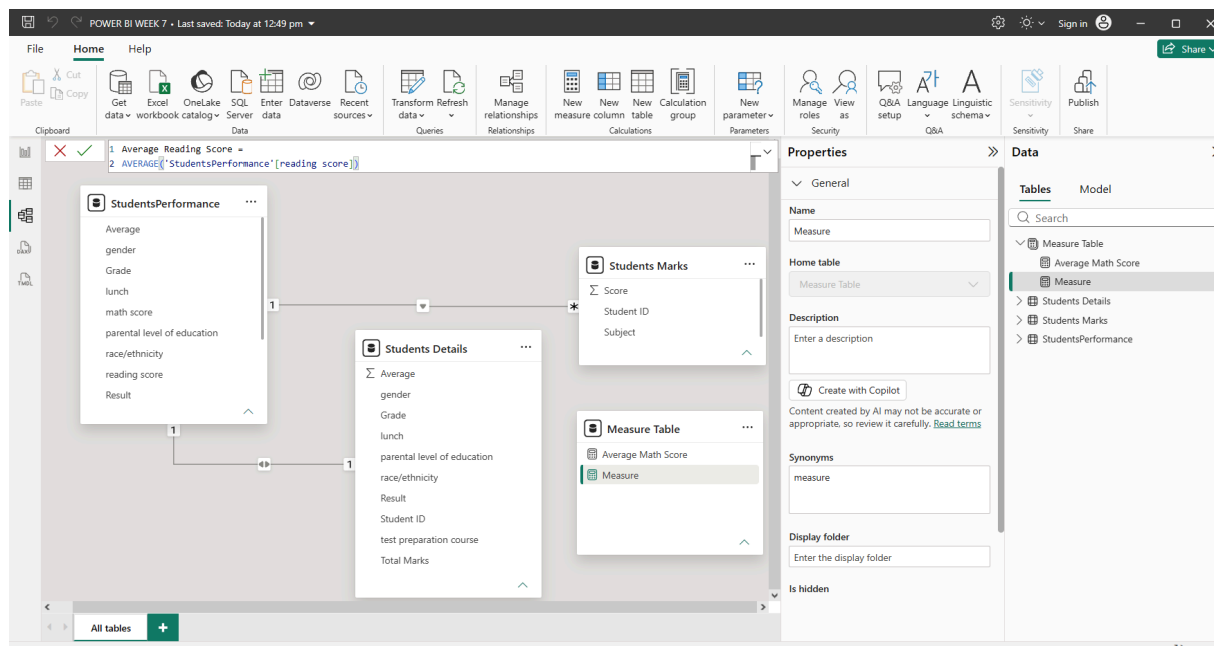
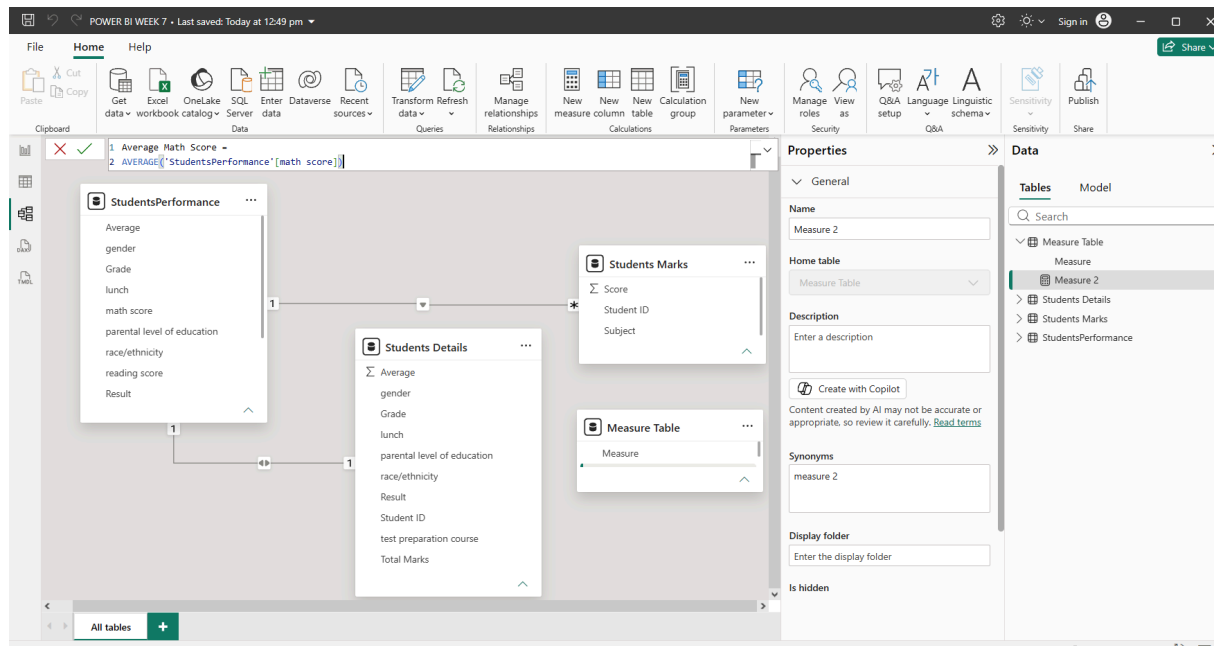
DISTINCTCOUNT('Students Details'[Student id]),

0

)

These measures are used to calculate subject-wise averages, overall average performance, and the percentage of students who passed.

Screenshot: Five DAX measures created in the Measure Table.



POWER BI WEEK 7 - Last saved: Today at 12:49 pm

File Home Help

Clipboard

Get data - workbook catalog - Data

OneLake SQL Server Data

Datastage Recent sources

Transform Refresh data - Queries

Manage relationships Relationships

New measure column Calculations

New table Calculation group

New parameter Parameters

Manage roles Security

View as Q&A Language Linguistic schema - Q&A

Sensitivity Publish

Share

1 Average Writing Score -
2 AVERAGE('StudentsPerformance'[writing score])

StudentsPerformance

- Average
- gender
- Grade
- lunch
- math score
- parental level of education
- race/ethnicity
- reading score
- Result

Students Details

- Average
- gender
- Grade
- lunch
- parental level of education
- race/ethnicity
- Result
- Student ID
- test preparation course
- Total Marks

Students Marks

- Score
- Student ID
- Subject

Measure Table

- Average Math Score
- Average Reading Score
- Measure

Properties

General

Name

Measure

Home table

Measure Table

Description

Enter a description

Create with Copilot

Content created by AI may not be accurate or appropriate, so review it carefully. [Read terms](#)

Synonyms

measure

Display folder

Enter the display folder

Is hidden

Data

Tables Model

Search

Measure Table

- Average Math Score
- Average Reading Score
- Average Writing Score
- Measure

Students Details

- Average
- gender
- Grade
- lunch
- parental level of education
- race/ethnicity
- Result
- Student ID
- test preparation course
- Total Marks

StudentsPerformance

All tables +

POWER BI WEEK 7 - Last saved: Today at 12:49 pm

File Home Help

Clipboard

Get data - workbook catalog - Data

OneLake SQL Server Data

Datastage Recent sources

Transform Refresh data - Queries

Manage relationships Relationships

New measure column Calculations

New table Calculation group

New parameter Parameters

Manage roles Security

View as Q&A Language Linguistic schema - Q&A

Sensitivity Publish

Share

1 Overall Average Score -
2 AVERAGE('StudentsDetails'[Average])

StudentsPerformance

- Average
- gender
- Grade
- lunch
- math score
- parental level of education
- race/ethnicity
- reading score
- Result

Students Details

- Average
- gender
- Grade
- lunch
- parental level of education
- race/ethnicity
- Result
- Student ID
- test preparation course
- Total Marks

Students Marks

- Score
- Student ID
- Subject

Measure Table

- Average Math Score
- Average Reading Score
- Average Writing Score
- Overall Average Score

Properties

General

Name

Overall Average Score

Home table

Measure Table

Description

Enter a description

Create with Copilot

Content created by AI may not be accurate or appropriate, so review it carefully. [Read terms](#)

Synonyms

overall average score, average score

Display folder

Enter the display folder

Is hidden

Data

Tables Model

Search

Measure Table

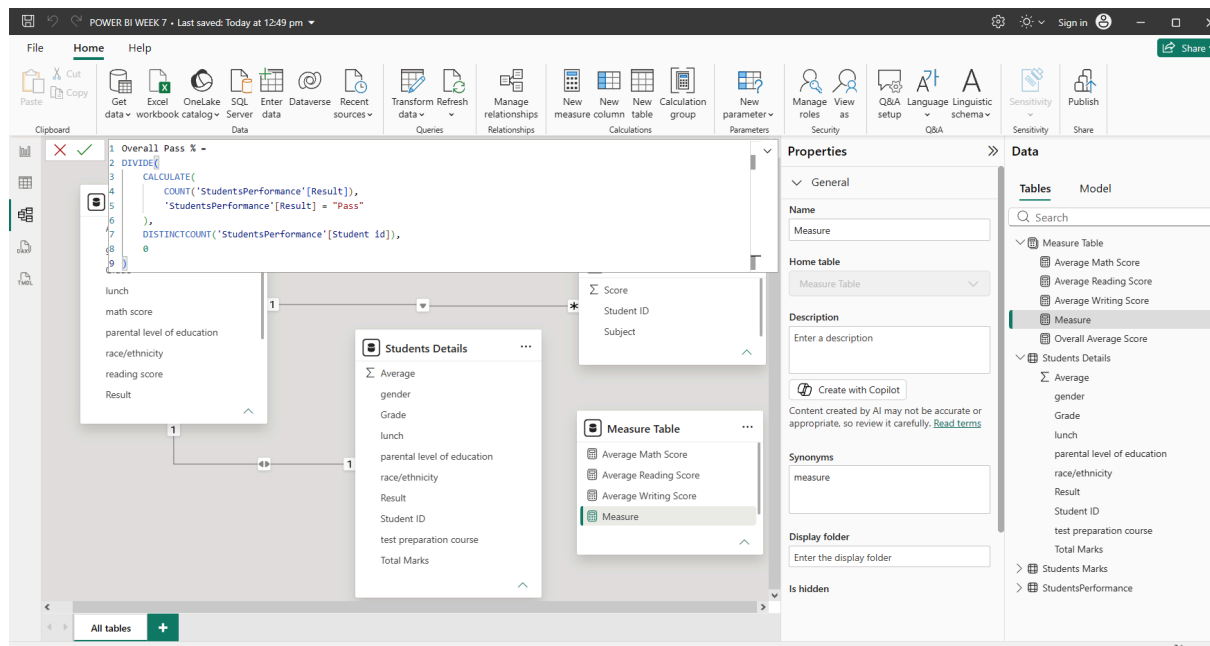
- Average Math Score
- Average Reading Score
- Average Writing Score
- Overall Average Score

Students Details

- Average
- gender
- Grade
- lunch
- parental level of education
- race/ethnicity
- Result
- Student ID
- test preparation course
- Total Marks

StudentsPerformance

All tables +



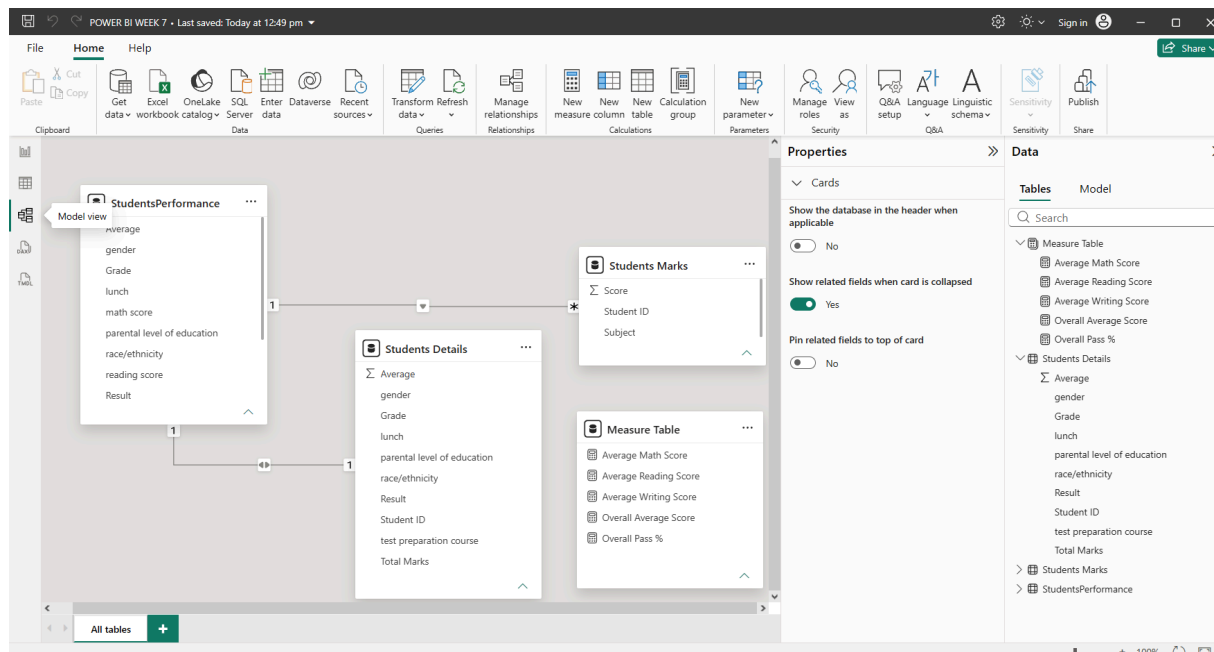
STEP 13 – VALIDATE THE RESULTS

Finally, validate the completed Power BI model.

1. Check the **Total marks** data type.
2. Check the **Average** values.
3. Check the **Grade** and **Result** columns.
4. Check the **Students Mark** table.
5. Check the relationships in **Model View**.
6. Check that **Overall Pass %** is formatted as a percentage.

The validation helps ensure that the prepared data and calculations are ready for further analysis. Week7_Report

Screenshot: Final validated model and measures.



CONCLUSION

The **Students Performance dataset** was successfully cleaned, transformed, and prepared for analysis using Power BI. A unique Student ID, Total Marks, Average, Grade, and Result were created. The subject scores were unpivoted into the **Students Mark** table, relationships were established between the tables, and DAX measures were created for academic performance analysis.